

Ports & Security-Group Rules

Parking Reservation System · Huawei Cloud Stack (CCE / OBS) deployment · May 5, 2026

The rules below are grouped by traffic direction. Only the highlighted entries are application-specific — the rest are standard CCE plumbing that your cloud-admin team likely already has in place.

1. North–South (users → app)

From	To	Port	Notes
End users (Internet / corp LAN)	HCS ELB <i>or</i> ingress controller	443/TCP	HTTPS. Required.
End users	Same	80/TCP	Optional — only if you want HTTP → HTTPS redirect at the edge.
HCS ELB → CCE worker nodes (<i>Option B only, NodePort</i>)	Worker node IPs	30080/TCP	Matches <code>nodePort: 30080</code> in <code>ingress-clusterip.yaml</code> . Open only on the worker subnet, not to the internet.
HCS ELB → CCE ingress controller pods (<i>Option A</i>)	Cluster ingress LB	80/TCP, 443/TCP	Whatever your Nginx Ingress LoadBalancer Service exposes.

2. East–West (inside the cluster)

Pod-to-pod traffic on the CCE pod network. Doesn't need cloud-level security-group rules — but if you have **Kubernetes NetworkPolicies** enabled, allow:

From	To	Port
parking-frontend pods	parking-backend Service	8001/TCP
Ingress controller pods	parking-frontend Service	80/TCP
Ingress controller pods	parking-backend Service	8001/TCP

3. Egress from CCE worker nodes

This is where most security-group work happens on HCS:

From	To	Port	Why
CCE worker subnet	Couchbase Enterprise cluster	11207/TCP	TLS data (KV) — primary SDK traffic. Required.
CCE worker subnet	Couchbase Enterprise cluster	18091/TCP	TLS management / bootstrap — required by the SDK on connect.
CCE worker subnet	Couchbase Enterprise cluster	18093/TCP	TLS Query (N1QL) — required; the app uses Couchbase queries.

From	To	Port	Why
CCE worker subnet	Couchbase Enterprise cluster	18094/TCP	TLS Search (FTS) — only if you enable full-text search (not used today; safe to omit).
CCE worker subnet	OBS endpoint	443/TCP	HTTPS S3 API for floor-layout uploads. Required.
CCE worker subnet	SWR registry	443/TCP	Image pull on rollout. Required.
CCE worker subnet	DNS resolvers	53/UDP, 53/TCP	Standard.
CCE worker subnet	(optional) Emergent LLM API	443/TCP	Only if <code>EMERGENT_LLM_KEY</code> is set and AI insights are used. Public-internet egress.

4. Operations plane

From	To	Port
Ops workstation	CCE Kubernetes API	5443/TCP (HCS CCE default; vanilla K8s is 6443)
Build host	SWR	443/TCP

5. CCE-internal (managed by HCS — listed for completeness)

Your CCE cluster will already have these between control plane and workers — don't tamper with them:

- 10250/TCP (kubelet)
- 10256/TCP (kube-proxy health)
- 30000–32767/TCP (NodePort range)
- VXLAN/Geneve overlay UDP (4789 / 6081)

Quick checklist for the customer's Sec-Group team

The **non-default** rules they need to add for this app are:

- ☐ Allow **CCE worker subnet** → **Couchbase cluster** on 11207, 18091, 18093/TCP .
- ☐ Allow **CCE worker subnet** → **OBS endpoint** on 443/TCP .
- ☐ Allow **CCE worker subnet** → **SWR** on 443/TCP .
- ☐ Allow **HCS ELB** → **CCE workers** on 443/TCP (Option A) or 30080/TCP (Option B).
- ☐ Allow **end users** → **HCS ELB** on 443/TCP .

Everything else is either intra-cluster (Kubernetes handles it) or already-open standard plumbing.